SplitFlow

A Bit-Shift product

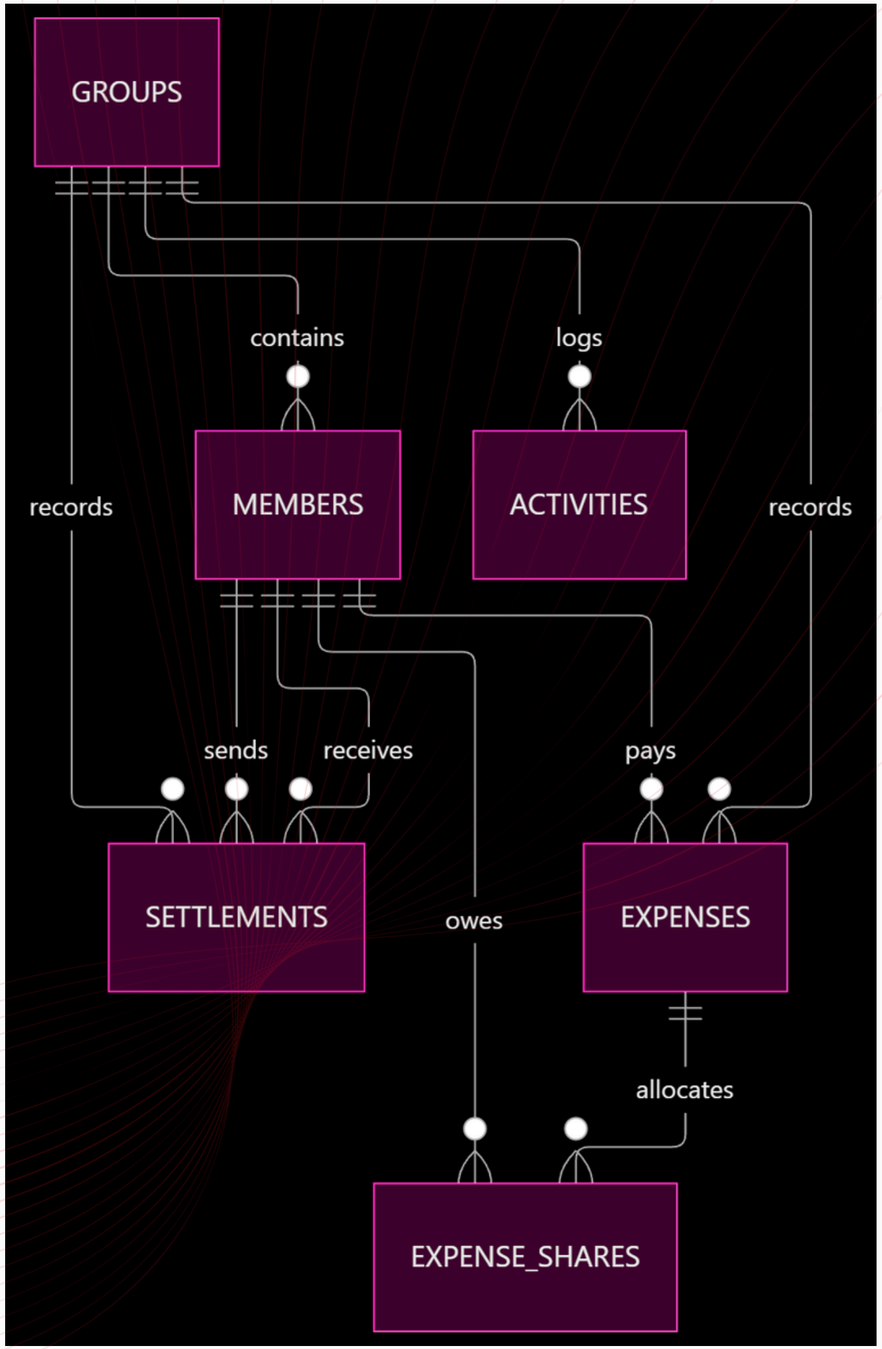
Solving the Group Expense Chaos

Simplifying how friends, roommates, and travel groups manage shared costs

01 **The Problem**

02 **The Dinner Scenario**

03 **The Solution**



Financial Logic and Split Methods

Accurate arithmetic using integer paise to eliminate rounding errors

Visual Insight: Key Graphic Highlight

Person	Paid	Share	Net Balance
Aarav	₹1,200	₹300	+₹900
Diya	₹0	₹300	-₹300
Kabir	₹0	₹300	-₹300
Meera	₹0	₹300	-₹300
Note	Payer	has positive	balance

Supported Split Methods

Equal split: deterministic paise distribution. Percentage: user-defined 100% shares. Weighted: proportional weights. Exact: matches total expense.

+

Debt Simplification Algorithm

Greedy: matches largest debt/credit. Transaction Reduction: minimizes repayments. Recommendation Engine: suggests moves without auto-executing funds.

Full-Stack Technical Architecture

A modular 3-layer system built for reliability and financial precision

1

Frontend

VUE 3

Backend

FastAPI

2

3

Database

SQLite

Relational Data Schema

Structured backend storage for comprehensive group activity tracking

Table	Purpose
groups	Name, description, and optional budget
members	Group membership and user roles
expenses	Amount, payer, category, and split method
expense_shares	Individual shares (recorded in paise)
settlements	Recorded repayments between members
activities	Audit log of group history messages

1

Paise Precision

Values like ₹125.50 stored as 12550 paise to prevent math drift

2

CORS Security

Configured to allow local development origins (port 5173)

Roadmap and Future Vision

From current MVP status to a production-ready financial ecosystem

01

Current MVP Features

Enables expense management, settlement planning, and CSV export.

02

AI Integration

Adds natural-language queries and automated spending categorization.

03

Payment Verification

Connecting to real-time gateways to confirm UPI transaction status.

04

Production Security

Implementing server-side auth and per-user authorization logic.